SN74ALVCH162830
1-BIT TO 2-BIT ADDRESS DRIVER WITH 3-STATE OUTPUTS

SCES082I–AUGUST 1996–REVISED JULY 2004

FEATURES

- **Member of the Texas Instruments Widebus™ Family**
- **Output Ports Have Equivalent 26-Ω Series Resistors, So No External Resistors Are Required**
- **Bus Hold on Data Inputs Eliminates the Need for External Pullup/Pulldown Resistors**
- **Latch-Up Performance Exceeds 250 mA Per JESD 17**
- **ESD Protection Exceeds JESD 22**
  - **2000-V Human-Body Model (A114-A)**
  - **200-V Machine Model (A115-A)**
  - **1000-V Charged-Device Model (C101)**

DESCRIPTION/ORDERING INFORMATION

This 1-bit to 2-bit address driver is designed for 1.65-V to 3.6-V V_{CC} operation.

Active bus-hold circuitry holds unused or undriven inputs at a valid logic state. Use of pullup or pulldown resistors with the bus-hold circuitry is not recommended.

The outputs, which are designed to sink up to 12 mA, include equivalent 26-Ω resistors to reduce overshoot and undershoot.

To ensure the high-impedance state during power up or power down, $\overline{OE}$ should be tied to V_{CC} through a pullup resistor; the minimum value of the resistor is determined by the current-sinking capability of the driver.

DBB PACKAGE
(TOP VIEW)

2Y2	1	80	1Y3
1Y2	2	79	2Y3
GND	3	78	GND
2Y1	4	77	1Y4
1Y1	5	76	2Y4
V_{CC}	6	75	V_{CC}
A1	7	74	1Y5
A2	8	73	2Y5
GND	9	72	GND
A3	10	71	1Y6
A4	11	70	2Y6
GND	12	69	GND
A5	13	68	1Y7
A6	14	67	2Y7
V_{CC}	15	66	V_{CC}
A7	16	65	1Y8
A8	17	64	2Y8
GND	18	63	GND
A9	19	62	1Y9
$\overline{OE1}$	20	61	2Y9
$\overline{OE2}$	21	60	1Y10
A10	22	59	2Y10
GND	23	58	GND
A11	24	57	1Y11
A12	25	56	2Y11
V_{CC}	26	55	V_{CC}
A13	27	54	1Y12
A14	28	53	2Y12
GND	29	52	GND
A15	30	51	1Y13
A16	31	50	2Y13
GND	32	49	GND
A17	33	48	1Y14
A18	34	47	2Y14
V_{CC}	35	46	V_{CC}
2Y18	36	45	1Y15
1Y18	37	44	2Y15
GND	38	43	GND
2Y17	39	42	1Y16
1Y17	40	41	2Y16

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PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of the Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

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ORDERING INFORMATION

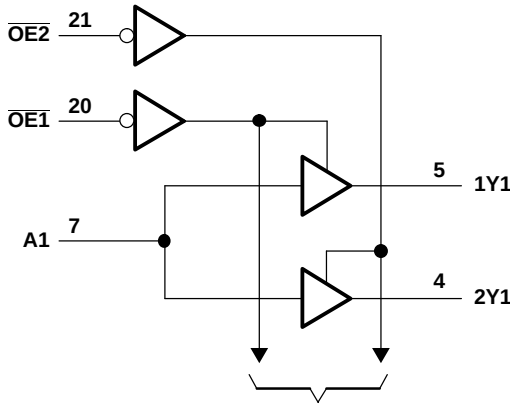
T_A	PACKAGE(1)		ORDERABLE PART NUMBER	TOP-SIDE MARKING
-40°C to 85°C	TVSOP - DBB	Tape and reel	SN74ALVCH162830GR	ALVCH162830

(1) Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.

FUNCTION TABLE

INPUTS			OUTPUTS	
$\overline{OE1}$	$\overline{OE2}$	A	1Yn	2Yn
L	H	H	H	Z
L	H	L	L	Z
H	L	H	Z	H
H	L	L	Z	L
L	L	H	H	H
L	L	L	L	L
H	H	X	Z	Z

LOGIC DIAGRAM (POSITIVE LOGIC)



To 17 Other Channels

ABSOLUTE MAXIMUM RATINGS(1)

over operating free-air temperature range (unless otherwise noted)

			MIN	MAX	UNIT
V_{CC}	Supply voltage range		-0.5	4.6	V
V_I	Input voltage range(2)		-0.5	4.6	V
V_O	Output voltage range(2)(3)		-0.5	V_{CC} + 0.5	V
I_{IK}	Input clamp current	V_I < 0		-50	mA
I_{OK}	Output clamp current	V_O < 0		-50	mA
I_O	Continuous output current			±50	mA
	Continuous current through each V_{CC} or GND			±100	mA
θ_{JA}	Package thermal impedance(4)			64	°C/W
T_{stg}	Storage temperature range		-65	150	°C

(1) Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input negative-voltage and output voltage ratings may be exceeded if the input and output current ratings are observed.
(3) This value is limited to 4.6 V, maximum.
(4) The package thermal impedance is calculated in accordance with JESD 51-7.

RECOMMENDED OPERATING CONDITIONS(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage		1.65	3.6	V
V_{IH}	High-level input voltage	V_{CC} = 1.65 V to 1.95 V	0.65 × V_{CC}		V
		V_{CC} = 2.3 V to 2.7 V	1.7		
		V_{CC} = 2.7 V to 3.6 V	2		
V_{IL}	Low-level input voltage	V_{CC} = 1.65 V to 1.95 V		0.35 × V_{CC}	V
		V_{CC} = 2.3 V to 2.7 V		0.7	
		V_{CC} = 2.7 V to 3.6 V		0.8	
V_I	Input voltage		0	V_{CC}	V
V_O	Output voltage		0	V_{CC}	V
I_{OH}	High-level output current	V_{CC} = 1.65 V		-2	mA
		V_{CC} = 2.3 V		-6	
		V_{CC} = 2.7 V		-8	
		V_{CC} = 3 V		-12	
I_{OL}	Low-level output current	V_{CC} = 1.65 V		2	mA
		V_{CC} = 2.3 V		6	
		V_{CC} = 2.7 V		8	
		V_{CC} = 3 V		12	
$\Delta t/\Delta v$	Input transition rise or fall rate			10	ns/V
T_A	Operating free-air temperature		-40	85	°C

(1) All unused control inputs of the device must be held at V_{CC} or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.

ELECTRICAL CHARACTERISTICS

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	V_{CC}	MIN	TYP(1)	MAX	UNIT
V_{OH}		I_{OH} = -100 µA	1.65 V to 3.6 V	V_{CC} - 0.2			V
		I_{OH} = -2 mA	1.65 V	1.2			
		I_{OH} = -4 mA	2.3 V	1.9			
		I_{OH} = -6 mA	2.3 V	1.7			
			3 V	2.4			
		I_{OH} = -8 mA	2.7 V	2			
		I_{OH} = -12 mA	3 V	2			
V_{OL}		I_{OL} = 100 µA	1.65 V to 3.6 V			0.2	V
		I_{OL} = 2 mA	1.65 V			0.45	
		I_{OL} = 4 mA	2.3 V			0.4	
		I_{OL} = 6 mA	2.3 V			0.55	
			3 V			0.55	
		I_{OL} = 8 mA	2.7 V			0.6	
		I_{OL} = 12 mA	3 V			0.8	
I_I		V_I = V_{CC} or GND	3.6 V			±5	µA
$I_{I(hold)}$		V_I = 0.58 V	1.65 V	25			µA
		V_I = 1.07 V	1.65 V	-25			
		V_I = 0.7 V	2.3 V	45			
		V_I = 1.7 V	2.3 V	-45			
		V_I = 0.8 V	3 V	75			
		V_I = 2 V	3 V	-75			
		V_I = 0 to 3.6 V(2)	3.6 V			±500	
I_{OZ}		V_O = V_{CC} or GND	3.6 V			±10	µA
I_{CC}		V_I = V_{CC} or GND, I_O = 0	3.6 V			40	µA
ΔI_{CC}		One input at V_{CC} - 0.6 V, Other inputs at V_{CC} or GND	3 V to 3.6 V			750	µA
C_i	Control inputs	V_I = V_{CC} or GND	3.3 V		4.5		pF
	Data Inputs				5		
C_o	Outputs	V_O = V_{CC} or GND	3.3 V		7.5		pF

(1) All typical values are at V_{CC} = 3.3 V, T_A = 25°C.
(2) This is the bus-hold maximum dynamic current. It is the minimum overdrive current required to switch the input from one state to another.

SWITCHING CHARACTERISTICS

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CC} = 1.8 V	V_{CC} = 2.5 V ± 0.2 V		V_{CC} = 2.7 V		V_{CC} = 3.3 V ± 0.3 V		UNIT
			TYP	MIN	MAX	MIN	MAX	MIN	MAX	
t_{pd}	A	Y	(1)	1.2	3.8		4	1.7	3.5	ns
t_{en}	$\overline{OE}$	Y	(1)	1	5.7		5.7	1	4.8	ns
t_{dis}	$\overline{OE}$	Y	(1)	1.5	6.2		5.4	1.7	5.2	ns

(1) This information was not available at the time of publication.

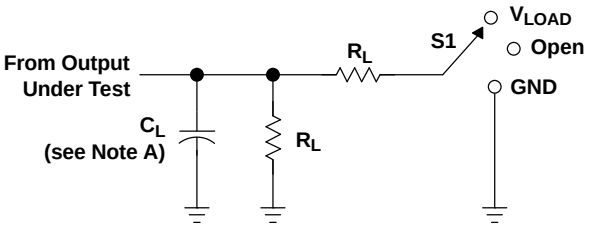
OPERATING CHARACTERISTICS

T_A = 25°C

PARAMETER		TEST CONDITIONS	V_{CC} = 1.8 V	V_{CC} = 2.5 V	V_{CC} = 3.3 V	UNIT
			TYP	TYP	TYP	
C_{pd} Power dissipation capacitance per bit (two outputs switching)	All outputs enabled	C_L = 0 pF, f = 10 MHz	(1)	50	54	pF
	All outputs disabled		(1)	8	8	

(1) This information was not available at the time of publication.

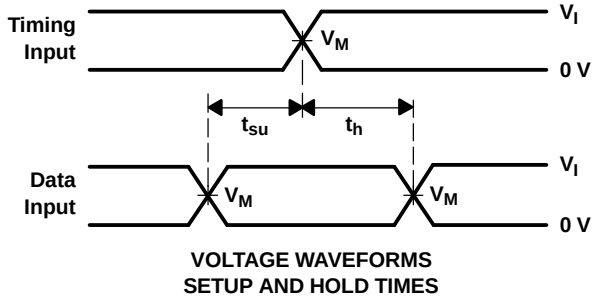
PARAMETER MEASUREMENT INFORMATION



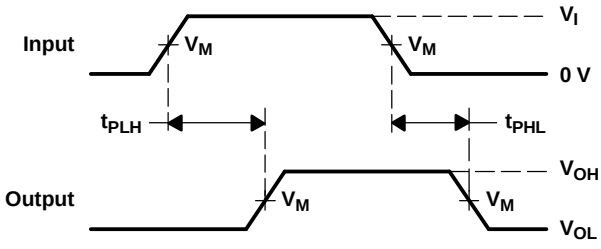
LOAD CIRCUIT

TEST	S1
t_{pd}	Open
t_{PLZ}/t_{PZL}	V_{LOAD}
t_{PHZ}/t_{PZH}	GND

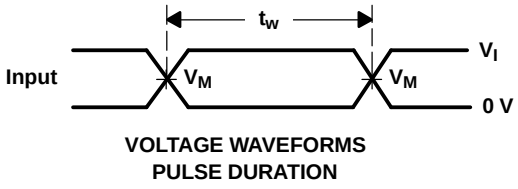
V_{CC}	INPUT		V_M	V_{LOAD}	C_L	R_L	V_Δ
	V_I	t_r/t_f					
1.8 V	V_{CC}	≤2 ns	$V_{CC}/2$	$2 \times V_{CC}$	30 pF	1 kΩ	0.15 V
2.5 V ± 0.2 V	V_{CC}	≤2 ns	$V_{CC}/2$	$2 \times V_{CC}$	30 pF	500 Ω	0.15 V
2.7 V	2.7 V	≤2.5 ns	1.5 V	6 V	50 pF	500 Ω	0.3 V
3.3 V ± 0.3 V	2.7 V	≤2.5 ns	1.5 V	6 V	50 pF	500 Ω	0.3 V



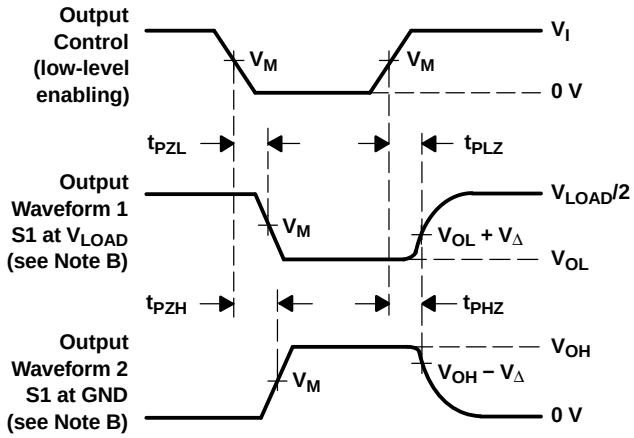
VOLTAGE WAVEFORMS
SETUP AND HOLD TIMES



VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES



VOLTAGE WAVEFORMS
PULSE DURATION



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES

NOTES: A. C_L includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low, except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
C. All input pulses are supplied by generators having the following characteristics: PRR ≤ 10 MHz, Z_O = 50 Ω.
D. The outputs are measured one at a time, with one transition per measurement.
E. t_{PLZ} and t_{PHZ} are the same as t_{dis}.
F. t_{PZL} and t_{PZH} are the same as t_{en}.
G. t_{PLH} and t_{PHL} are the same as t_{pd}.
H. All parameters and waveforms are not applicable to all devices.

Figure 1. Load Circuit and Voltage Waveforms

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74ALVCH162830GR	Active	Production	TSSOP (DBB) | 80	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	ALVCH162830
SN74ALVCH162830GR.B	Active	Production	TSSOP (DBB) | 80	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	ALVCH162830

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

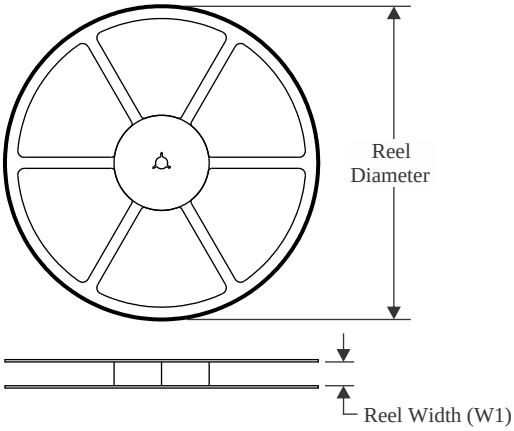
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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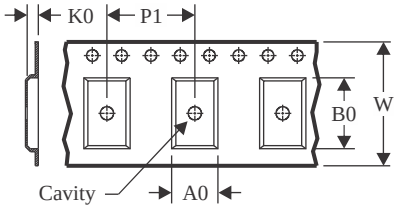
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TAPE AND REEL INFORMATION

REEL DIMENSIONS

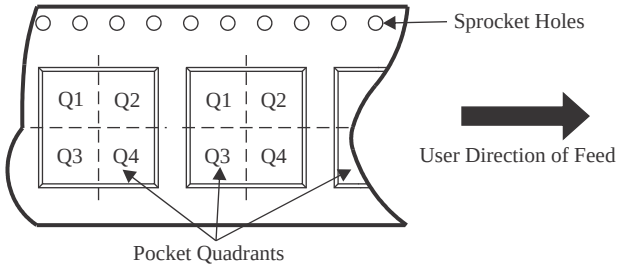


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

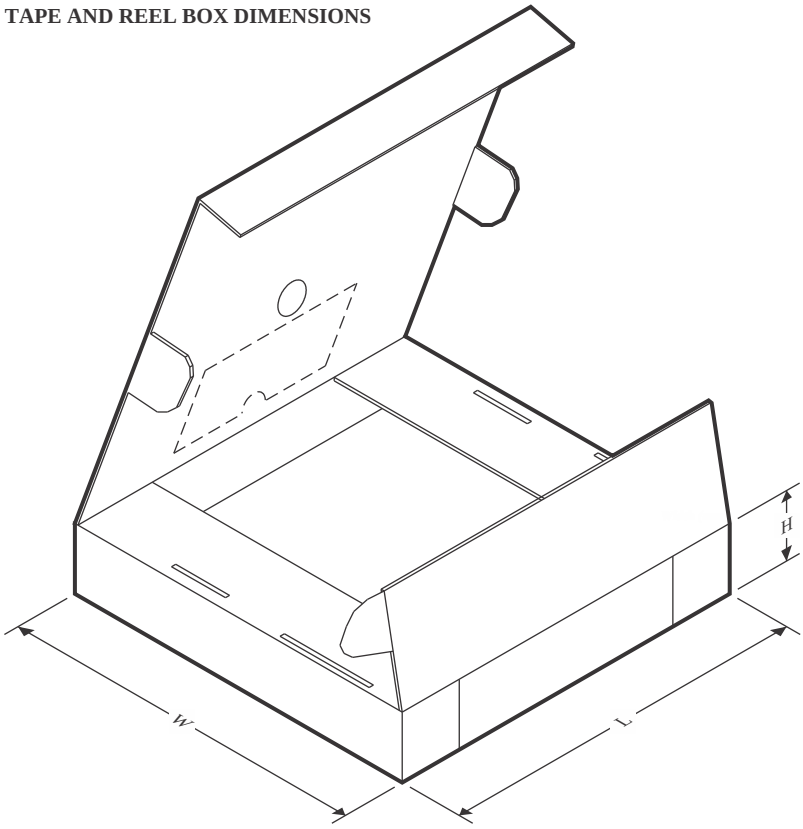
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

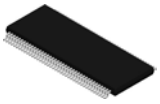
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74ALVCH162830GR	TSSOP	DBB	80	2000	330.0	24.4	8.4	17.3	1.7	12.0	24.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74ALVCH162830GR	TSSOP	DBB	80	2000	356.0	356.0	45.0

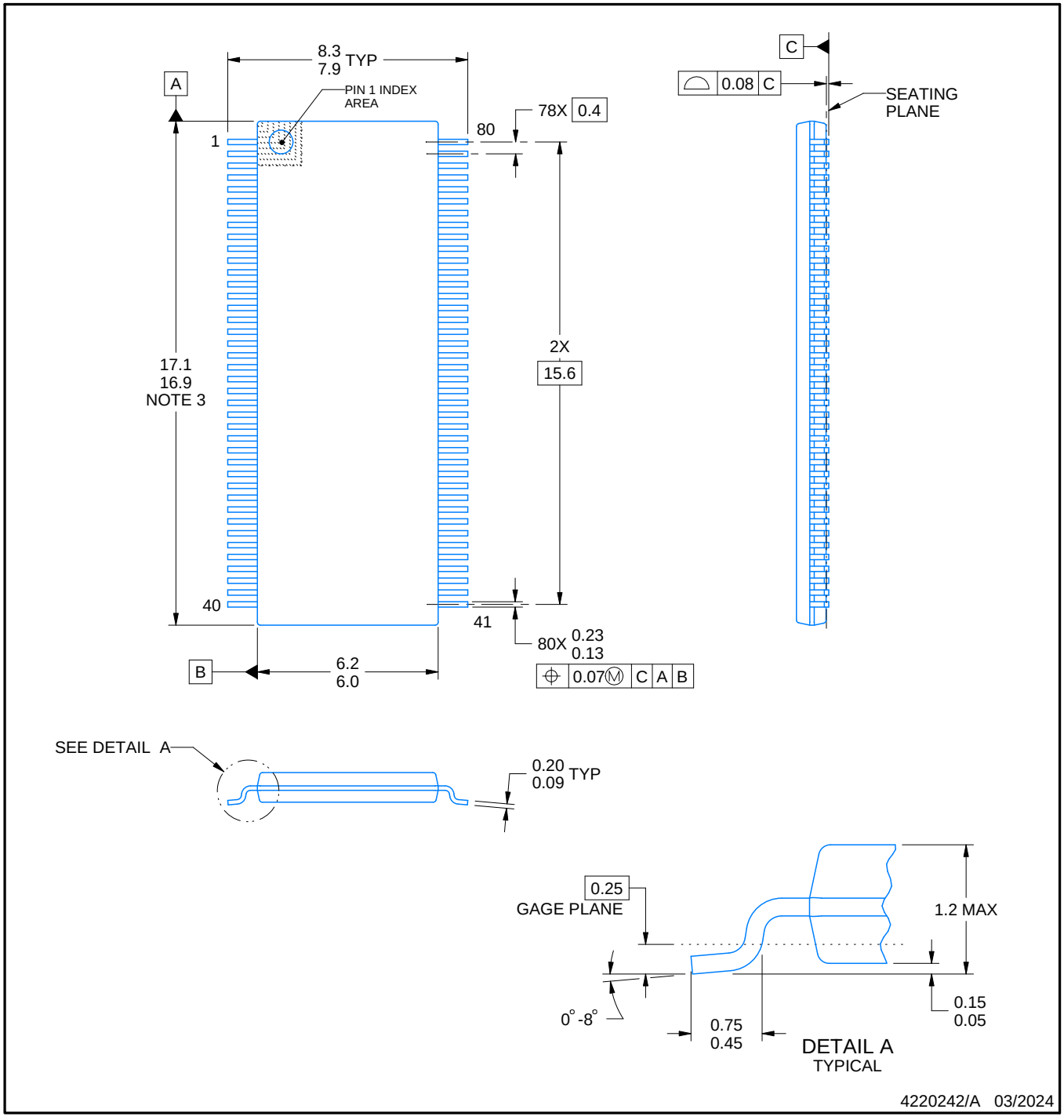


PACKAGE OUTLINE

DBB0080A

TVSOP - 1.2 mm max height

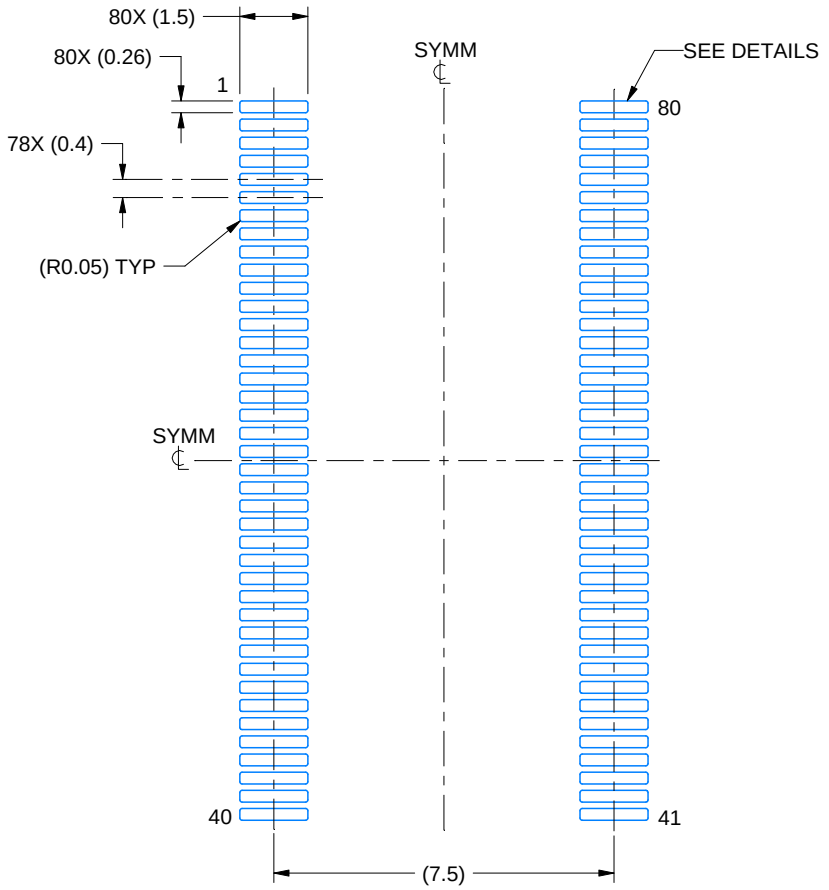
SMALL OUTLINE PACKAGE



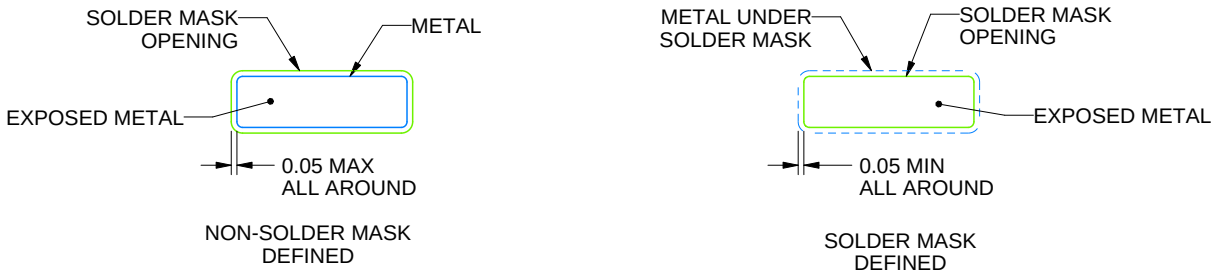
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NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-153, Variation FF.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 6X

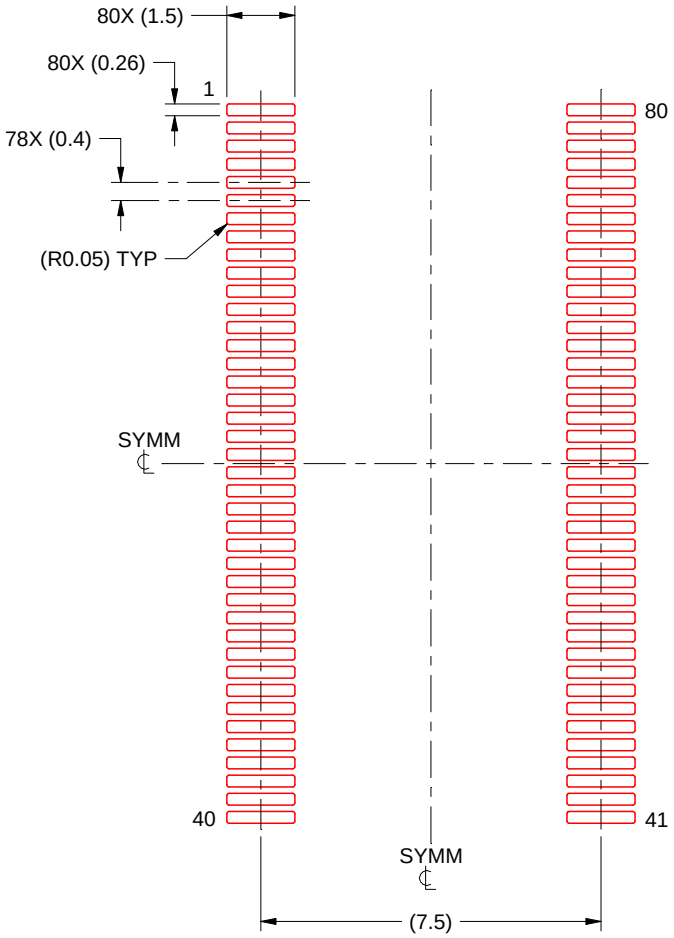


SOLDER MASK DETAILS

4220242/A 03/2024

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
7. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature numbers SLMA002 (www.ti.com/lit/slma002) and SLMA004 (www.ti.com/lit/slma004).
8. Size of metal pad may vary due to creepage requirement.
9. Vias are optional depending on application, refer to device data sheet. It is recommended that vias under paste be filled, plugged or tented.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 6X

4220242/A 03/2024

NOTES: (continued)

10. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
11. Board assembly site may have different recommendations for stencil design.

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